

MOTIONCORE · CALCULATION NOTE

MotionCore sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-02	Amish Chadha	Decisions of 2026-10-02 noted under R9, R10, R11 and R13; figures unchanged until sizing.py includes the economizer
0.3	2026-10-01	Amish Chadha	Constructable design (MTC-DDR-003). Finned tube on a floor plate with a sealant joint (new floor joint estimate), 3 mm full-height fins, envelope, mass and cost updated; budget reported as a value-engineering target
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002). R1 upper bound 58 to 60 V (CellGuard 16S), controller overvoltage fault 60 to 66 V, fuse rating 60 V DC or more, R12 mass limit 1.5 to 2.0 kg
0.1	2026-09-25	Amish Chadha	First TRL 3 sizing note (currents, fuses and leads, thermal, e-stop timing, precharge, speed sensing, reaction times, CAN and bus voltage, size, mass, cost) against the TRL 3 decisions in MTC-DDR-001

MotionCore sizing calculations

On paper, MotionCore meets ten of its fourteen requirements and none is outright not met. R12 is met after Amish relaxed the mass limit from 1.5 kg to 2.0 kg (MTC-DDR-002): the constructable design of MTC-DDR-003 weighs about 1.91 kg, a margin of 0.09 kg, because the finned enclosure that is also the heat sink weighs about 1.0 kg on its own. R1 now runs to 60 V so that CellGuard's 16S LiFePO4 packs (58.4 V full) are covered. R2 and R9 are **at risk** in the heavy case: at 350 W on an 8S LiFePO4 pack the walls reach about 58.4 °C and the floor plate about 60.0 °C against 60 °C, and reasonable changes to the controller assumptions push it to 65 to 71 °C. R10 is **at risk** because the XT90 power socket is not sealed, and R11 (fit time) cannot be verified at TRL 3. The reference case is comfortable: about 50 °C on the case, 6.9 A from a SwapCell pack, e-stop power removal within 67 ms worst case, and an estimated MotionCore kit cost of USD 293 against a value-engineering target of USD 300 (USD 7 under), with the USD 70 reference motor costed to each host.

Every number here is printed by `docs/04-calcs/sizing.py` (run from the repo root: `python docs/04-calcs/sizing.py`), which also writes `docs/04-calcs/results.csv`. The script reads the enclosure dimensions from `cad/src/model.py` and the costs from `bom/bom.csv`. All values are first-principles estimates; nothing is measured.

■ 1. Assumptions

Table 1. Inputs. All are assumptions for a paper design; part-specific values are to be confirmed from datasheets once parts are chosen.

Input	Value	Basis
Reference pack	SwapCell 13S lithium-ion, 46.8 V nominal, 39.0 to 54.6 V, 110 mΩ, 20 A continuous, 35 A for 10 s	SwapCell interface v0.3 (SWC-PRC-001 v0.3)
CargoMule pack	12S LiFePO ₄ , 38.4 V nominal, 30.0 to 43.8 V, about 60 mΩ, 15 A host limit	CargoMule precis (CGM-PRC-001); resistance assumed
Heavy-case pack	8S LiFePO ₄ , 25.6 V nominal, 20.0 to 29.2 V, about 30 mΩ	PalletPilot and StepClimber class; resistance assumed
Motor efficiency	80 % at rated load, 75 % at 750 W for 10 s	Typical small geared hub
Phase to bus current ratio	1.2 (winding matched to the host's speed)	Assumption; 2.0 checked as a sensitivity
Controller losses	3.9 mΩ per phase (MOSFET 2.4 mΩ hot, shunt 0.5 mΩ, copper 1.0 mΩ); 100 ns switching edges at 30 kHz; 1.5 W fixed	Typical VESC-class 75 V stage
Power path in the module	Contactors 0.5 mΩ, internal leads 1.5 mΩ, XT90 pair 0.6 mΩ, fuse 3.5 mΩ (20 A) or 1.5 mΩ (40 A)	Typical parts
Auxiliary supply	Coil 4.0 W at 12 V, supervisor 0.6 W, buck efficiency 85 %	Typical sealed 100 A contactor without economizer
Enclosure	220 x 140 mm finned tube 56 mm tall on a 3 mm floor plate (59 mm body), 3 mm walls, 3 mm lid, 22 fins 3 x 14 x 56 mm; clear anodized, ε 0.80, solar absorptance 0.35	cad/src/model.py (MTC-DDR-003)
Floor joint	Thermally conductive silicone sealant, 1.0 W/m·K, 0.2 mm film over 2,380 mm ² of wall and port end faces	Assumed; to confirm with the chosen sealant
Convection	Vertical plates $h = 1.42 (\Delta T/L)^{0.25}$, lid $h = 1.32 (\Delta T/L)^{0.25}$, fin faces 0.8 of a free plate; underside ignored	Simplified laminar correlations for still air
CellGuard pack	16S LiFePO ₄ , 58.4 V full	R1 check only; pack resistance not known here
Contactors coil	36 Ω, 0.30 H, releases at 10 % of rated current; 10 ms armature travel; 5 ms arc	Assumed; release time 50 ms or less is a selection criterion
Bus capacitance	1,000 μF; precharge 100 Ω	To confirm from the chosen controller
Speed sensor	8 magnets; 2 % period jitter from magnet placement	Concept (MTC-PRC-001)
Ambient	40 °C, shaded	MTC-REQ-001 R9

■ 2. Operating points (R1, R2)

The controller is more efficient than the blanket 95 % assumed at TRL 2, so the pack supplies about 322 W, not 333 W, in the reference case. At the reference point about 5.7 W goes to the auxiliary supply and power path, 3.9 W to the controller and 62.5 W to motor and gear losses (Figure 4 in MTC-PRC-001).

Table 2. Operating points. Currents at nominal and minimum pack voltage.

Case	Pack power	Current, nominal V	Current, minimum V	Controller loss	Controller efficiency
Reference: 250 W, SwapCell	322 W	6.9 A at 46.8 V	8.3 A at 39.0 V	3.9 W	98.8 %
CargoMule: 250 W, 12S LFP	323 W	8.4 A at 38.4 V	10.8 A at 30.0 V	4.3 W	98.7 %
Heavy: 350 W, 8S LFP	453 W	17.7 A at 25.6 V	22.9 A at 20.0 V	9.0 W	98.0 %
500 W, 8S LFP (information)	648 W	25.3 A at 25.6 V	32.9 A at 20.0 V	15.5 W	97.6 %

For 750 W over 10 s at nominal voltage, the pack current is 21.9 A on SwapCell (inside its 35 A, 10 s limit) and 40.9 A on 8S LiFePO4. CargoMule's own 15 A pack limit caps its peak at about 414 W of shaft power: this is a host setting, which the supervisor respects, not a MotionCore limit. The supervisor sets the controller's battery current limit to the lower of the host setting and the pack's allowed discharge current in PACK_LIMITS.

All three host packs fall inside 20 to 60 V, and so does a full CellGuard 16S LiFePO4 pack at 58.4 V, 1.6 V below the new upper bound (MTC-DDR-002; it was 58 V). The controller is a 75 V class and the auxiliary buck accepts 18 to 75 V, so **R1 is met by design review**. R2 is met electrically; its heavy case depends on R9 and is **at risk**.

■ 3. Fuses, contactor and leads

The fuse must carry 1.25 times the continuous current at minimum pack voltage and hold the 10 s peak at no more than 110 % of its rating, which a blade fuse carries without opening.

Table 3. Fuse check.

Pack	Continuous at minimum V	10 s peak	Fuse needed	Fitted	Prospective short circuit
SwapCell	8.3 A	21.9 A	19.9 A	20 A	about 470 A
CargoMule	10.8 A	15.0 A (host limit)	13.6 A	20 A	about 663 A
8S LiFePO4	22.9 A	40.9 A	37.2 A	40 A	about 856 A

The 20 A and 40 A fuses proposed at TRL 2 hold, although the 20 A fuse has little margin for a SwapCell host that uses the full 750 W peak. The fuse must be rated for **60 V DC or more** (58 V before R1 was raised) with at least **1 kA** breaking capacity; common 32 V automotive fuses are not suitable. The prospective short-circuit current of a CellGuard 16S pack is not calculated here because its resistance is not known in this repo; it must be checked against the 1 kA breaking capacity before a CellGuard host is fitted. The contactor carries at most about 41 A for 10 s against its 100 A rating; the selection criterion that matters is breaking at least 50 A at 60 V DC.

Pack leads of 2.5 mm² over 1.5 m each way drop 0.14 V and lose 1.0 W in the reference case. In the heavy case at 20 V they drop 0.47 V and lose 10.8 W (3.1 % of shaft power); 4 mm² leads cut this to 0.30 V and 6.7 W, so 24 V hosts get 4 mm² pack leads (bom/bom.csv item 13).

■ 4. Thermal (R9)

The enclosure has 0.0308 m² of lid, 0.0446 m² of walls and 0.0382 m² of fin faces (0.0454 and 0.0338 m² before MTC-DDR-003: the fins now run the full tube height). At the heavy point the heat-transfer coefficients are about 6.5 W/m²K on the walls, 5.4 on the fins, 6.7 on the lid and 6.4 for radiation, giving a conductance of about 1.18 W/K (the TRL 2 estimate was 1.3 W/K). Heat is taken at minimum pack voltage, where the current is highest.

Table 4. Case and MOSFET temperatures at 40 °C ambient, shaded.

Case	Heat in the module	Case surface	MOSFET junction
Reference, 250 W on SwapCell	10.0 W	50.2 °C	about 52 °C
CargoMule, 250 W	11.2 W	51.1 °C	about 54 °C
Heavy, 350 W on 8S LFP	20.1 W	58.4 °C	about 67 °C
500 W on 8S LFP (information)	32.7 W	67.7 °C	about 84 °C

Floor joint (MTC-DDR-003). The controller, contactor and fuse block sit on the floor plate, which is a separate plate under the finned tube. Their heat (all of it except the supervisor's 0.7 W) crosses a 0.2 mm film of thermally conductive sealant on 2,380 mm² of wall and port end faces, a conductance of about 11.9 W/K. That puts the floor plate about 0.8 K above the walls in the reference case (51.0 °C) and 1.6 K above them in the heavy case (**60.0 °C**, at the R9 limit), and lifts the MOSFET junctions by the same amount (about 69 °C heavy). The floor plate's own underside is ignored, so these figures are on the safe side.

The derate to 350 W at 24 V (MTC-DDR-001 item 9) is needed: 500 W misses the 60 °C target by about 8 K. At 350 W the margin is only 1.5 K and depends on the controller assumptions:

- Phase current twice the bus current (a motor running well below its base speed, as a walk-behind host with a fast winding would): 37.2 W, **70.8 °C**.
- Twice the phase resistance (low-cost controller MOSFETs): 28.9 W, **65.0 °C**.
- Sun on the lid (information; R9 assumes shade): 10.8 W absorbed, 59.0 °C in the reference case and 66.4 °C in the heavy case.

The auxiliary supply is about 54 % of the reference-case heat, mostly the contactor coil. A coil economizer that holds the contactor at about 1 W (decided on 2026-10-02, wired so the safety loop breaks the coil supply upstream of it; not yet in this note's results or the BOM) would lower the heavy case to 16.5 W, 55.6 °C on the walls and about 57 °C on the floor plate (open decision 4 in MTC-DEC-001). **R9 is met in the reference case and at risk in the heavy case.**

■ 5. Emergency stop timing (R3)

Channel A of the e-stop opens the contactor coil circuit directly. With a 24 V Zener across the coil, the coil current decays from 333 mA to its release point in 2.9 ms ($\tau = 8.3$ ms); a plain diode takes 15.5 ms. Adding 10 ms for contact opening and bounce, 10 ms for armature travel, 5 ms of arc and 2.2 ms during which the bus capacitors (0.69 J) keep the motor running gives about **30 ms modeled** with the Zener and 43 ms with a plain diode.

The worst case uses the release time the contactor must meet as a selection criterion (50 ms or less with its suppressor): $10 + 50 + 5 + 2.2 \approx$ **67 ms**, inside the 100 ms of R3. In 67 ms a host at 25 km/h travels about 47 cm and a walk-behind machine at 1.5 m/s about 10 cm (69 cm and 15 cm in the full 100 ms). No firmware is in this

path, so R3 holds with both processors failed. **R3 is met on paper.** R4 (no automatic restart) is met by circuit review: the coil loop can close only after the key is cycled, and the supervisor closes its switch only at zero throttle with the brakes released and no fault.

■ 6. Precharge (R8)

With 100 Ω and 1,000 μF the time constant is 100 ms. At the 60 V upper bound of R1 the peak precharge current is 0.60 A and the residual at closure 0.15 V. The supervisor closes the contactor after six time constants (0.6 s) and only if the controller reports a bus voltage of at least 90 % of the pack voltage, which catches a shorted bus or a missing capacitor bank. The residual difference at closure is 0.07 to 0.14 V, and the closure inrush is 1.0 A (SwapCell), 1.2 A (CargoMule) and 1.3 A (8S LiFePO4), under the 5 A target. Peak precharge current is 0.55 A at 54.6 V, and the resistor absorbs 1.49 J per start. Into a shorted bus it would dissipate 30 W; a 1 s timeout limits that to 30 J, which a 10 W aluminum-clad resistor survives. **R8 is met.** The TRL 2 figure of 0.5 s (five time constants) left up to 8 A at closure on a low-resistance LiFePO4 pack, so the closure point moves to 0.6 s.

■ 7. Speed sensing and reaction times (R5 to R7)

Table 5. Speed sensor pulse rates with eight magnets.

Wheel	At 1.5 m/s	At 25 km/h
20 in, 1.57 m (CargoMule)	7.6 pulses/s (131 ms period)	35.4 pulses/s (28 ms)
700C or 28 in, 2.15 m (SunSpoke)	5.6 pulses/s (179 ms)	25.8 pulses/s (39 ms)
200 mm hub, 0.63 m (walk-behind)	19.0 pulses/s (52 ms)	88.2 pulses/s (11 ms)

Counting pulses over 0.5 s is too coarse at walking pace (one count is 26 % of the reading). The supervisor therefore times each period with a 1 MHz timer (resolution about 0.0035 % at 25 km/h); magnet placement jitter of about 2 % leaves a fivefold margin to the 10 % overspeed threshold. Overspeed held for 0.5 s opens the contactor in about 0.70 s at 1.5 m/s and 0.60 s at 25 km/h on a 20 in wheel, including one pulse period and the 67 ms contactor budget. **R6 is met on paper.** The wheel circumference and magnet count are configuration values; a wrong value moves the limit, so they belong on the fault chart.

Timing budgets: a brake switch sets the controller command to zero in about 75 ms (10 ms debounce, 5 ms loop, 10 ms CAN, 50 ms current ramp), inside 200 ms (**R5 met on paper**). A lost controller heartbeat takes about 170 ms (100 ms timeout plus loop, CAN and ramp), and the contactor is opened as well; a BMS fault, which SwapCell and CellGuard send on change, takes about 80 ms (**R7 met on paper**).

■ 8. CAN buses and bus voltage

The supervisor uses two CAN buses: an internal bus to the controller at 500 kbit/s (the VESC default) and an external bus to the SwapCell pack or CellGuard BMS at the SwapCell rate of 250 kbit/s. The internal bus carries five controller status frames at 50 Hz and commands at 100 Hz, about 11.2 % load. The external bus carries the pack's frames (1.8 %, SWC-CAL-001) plus the supervisor's HOST_HEARTBEAT and a status frame at 10 Hz each, about 2.9 % in total. Keeping the buses apart means the pack never sees controller traffic and no VESC setting has to change.

As a SwapCell host, the supervisor sends HOST_HEARTBEAT as host type 0 (vehicle) and requests mode 2 (discharge). The controller's battery regeneration current is set to zero unless the host enables regeneration, in which case the supervisor requests mode 4 with a host charge limit, as SwapCell interface v0.3 item C allows.

Regeneration at 10 A into a full SwapCell pack raises its terminals to about 55.9 V, inside 60 V. Raising R1 to 60 V means the controller's overvoltage fault can no longer sit at 60 V, where a full CellGuard 16S pack (58.4 V) would leave only 1.6 V of headroom; it moves to 66 V, 7.6 V above that pack and 6 V above the R1 bound. If the contactor opens while the controller is regenerating, the bus rises at about 10 V/ms, and with the fault at 66 V the energy left in the phase inductance (0.12 J) lifts the bus to about 67.8 V (62.0 V with the earlier 60 V setting), still inside the 75 V rating but with less margin. A direct-drive motor pushed faster than its no-load speed is the remaining risk: its back-EMF stays under 75 V only up to 1.25 times the no-load speed at 60 V (1.29 times at 58 V), so direct-drive hosts need a speed limit below that or a bus clamp.

■ 9. Size and mass (R12)

The module envelope of the constructable design (MTC-DDR-003) is 240 x 168 x 69 mm, inside 250 x 170 x 70 mm: the panel plate is gone, the lid sits on a 1 mm gasket with button-head screws, and the floor plate stands on 1 mm rivet nut flanges and 3 mm pads, so the tube is 56 mm tall to keep the total under 70 mm. The enclosure metal weighs about 1.00 kg (finned tube 0.50 kg, of which fins 0.14 kg; floor plate 0.25 kg; lid 0.25 kg), and the parts inside and on the walls about 0.91 kg (sockets, frame, speed socket and vent 0.14 kg; pads, rivet nuts, standoffs, screws, gasket and sealant 0.07 kg; supervisor on its wider carrier board 0.07 kg), so the module weighs about **1.91 kg against the 2.0 kg limit: R12 is met**, with 0.09 kg of margin (1.94 kg and 0.06 kg before MTC-DDR-003). Thinner fins (3 mm, typical for an extrusion) and the dropped panel plate more than offset the added fixings. Amish relaxed the limit from 1.5 kg to 2.0 kg (MTC-DDR-002) rather than thin the enclosure, because the enclosure is the heat sink and R9 is at risk. The kit weighs about 5.2 kg with the reference motor and 2.8 kg without it, not counting the bench rig.

■ 10. Cost (R14)

Value-engineering target: USD 300 (`budget_usd` , a hypothetical control target, not a limit). Estimated cost of the constructable design: **USD 293 for the MotionCore kit, items 2 to 14 of `bom/bom.csv` (USD 7 under the target)**. Making the design constructable added USD 28 (enclosure tube and floor plate, sockets with frames and flanges, the M8 speed socket and membrane vent, brake levers, the speed sensor plug, the harness branch, rivet nuts and fixings); it was USD 265 before MTC-DDR-003. The reference hub motor (item 1, USD 70) is costed to each host, which brings a complete kit with motor to USD 363. The bench rig (item 15, USD 35) belongs to the prototype only and is not part of the kit. All prices are indicative. **R14: USD 7 under the value-engineering target.**

■ 11. Results against requirements

Table 6. Requirement status (at risk first; none is not met).

ID	Value	Target	Status
R2	250 W: 6.9 A; 350 W: 17.7 A; 750 W for 10 s: 21.9 A (SwapCell), 40.9 A (8S LFP)	250 W reference; 350 W heavy; 750 W for 10 s	At risk (heavy case thermal, R9)
R9	50.2 °C reference; 58.4 °C heavy on the walls, 60.0 °C on the floor plate (65 to 71 °C in sensitivity cases)	60 °C at 40 °C ambient	At risk
R10	XT90 not sealed (bench prototype only, decided 2026-10-02); sealed power connector evaluated on paper against IP67 mated, 40 A or more at 60 V DC; M12 and motor plug sealed when mated; vibration not analyzed	IP65; survive vibration	At risk
R11	Interface v0.2 decided on 2026-10-02: v0.1 as drawn (MTC-DWG-001 Rev P3) plus the M8 speed sensor socket (MTC-DDR-003 A1); fit time needs a timed fit	One keyed set; fit in 2 h	Not verifiable at TRL 3
R1	Three host packs and CellGuard 16S (58.4 V) inside 20 to 60 V; 75 V controller; 18 to 75 V auxiliary buck	20 to 60 V; 75 V transients	Met (design review)
R3	67 ms worst case, 30 ms modeled; no firmware in the path	100 ms	Met (paper)
R4	Key reset; supervisor enable conditions	No automatic restart	Met (design review)
R5	75 ms	200 ms	Met (paper)
R6	Period method; 5x margin over jitter; trip 0.60 to 0.70 s	10 % for 0.5 s	Met (paper)
R7	170 ms worst	200 ms	Met (paper)
R8	0.55 A peak; 1.3 A at closure; ready in 0.6 s	Under 5 A; 1 s	Met
R13	MIT supervisor on its own processor; unmodified GPL-3.0 VESC firmware; CAN link only	Open hardware and firmware	Met (design review); written license review before any firmware is published
R12	240 x 168 x 69 mm; 1.91 kg	250 x 170 x 70 mm; 2.0 kg	Met (0.09 kg margin)
R14	USD 293 for items 2 to 14	Value-engineering target USD 300	USD 7 under the target (indicative prices)

■ 12. Checks against earlier documents

The TRL 2 figures in MTC-PRC-001 v0.2, MTC-REQ-001 v0.2 and the README were checked against this script and corrected in v0.3: pack power 333 W to 322 W; reference current 7.1 A to 6.9 A; CargoMule current 9.3 A at 36 V to 8.4 A at 38.4 V (CargoMule uses a 12S LiFePO4 pack); controller loss 16 W to 3.9 W; reference case temperature 55 °C to 50 °C; 500 W case 67 °C to 68 °C (now information only); heavy case now 350 W and 58.5 °C; e-stop time about 60 ms to 67 ms worst case; precharge closure 0.5 s to 0.6 s; module envelope 250 x 170 x 66 mm to 243 x 168 x 66 mm; module mass 1.3 kg to 1.94 kg; kit mass 4.4 kg to 5.3 kg; cost \$325 for the full kit to \$265 for the MotionCore kit plus \$70 for the reference motor (\$335 with motor). The speed-sensor pulse rates (7.5 and 35 pulses per second) and the precharge peak (0.55 A, 1.5 J) stand. In v0.2 (MTC-DDR-002): R1 upper bound 58 V to 60 V; overvoltage fault 60 V to 66 V and bus peak 62.0 V to 67.8 V; fuse rating 58 V DC to 60 V DC or more; direct-

drive back-EMF limit 1.29 to 1.25 times no-load speed; R12 limit 1.5 kg to 2.0 kg, status not met to met. In v0.3 (MTC-DDR-003): fins 4 mm to 3 mm and full height; envelope 243 x 168 x 66 mm to 240 x 168 x 69 mm; module mass 1.94 kg to 1.91 kg; heavy case 58.5 °C to 58.4 °C on the walls, with a new floor plate figure of 60.0 °C; kit cost \$265 to USD 293; R14 reported against the value-engineering target.

Safety: These are paper estimates for a module that switches a lithium pack able to deliver several hundred amperes into a short and drives moving machinery. They do not replace a circuit review, datasheet checks or testing of the safety functions. Nothing may be built or energized from this note; building and testing are TRL 4 work and on hold by Amish's instruction. The build plan MTC-BLD-001 carries the safety stops for the first build.